

School of Natural Sciences and Mathematics

Chemistry (BA, BS)

The Chemistry major builds on a base of chemistry, physics, mathematics, and computer science to provide the student the opportunity to develop essential theoretical and practical skills in the subdisciplines of organic, physical, inorganic, analytical, and macromolecular chemistry. Typically, the practice of chemistry in industry deals with the synthesis, analysis, and control of the many materials used in our technological society.

The Chemistry program at UT Dallas is designed to instruct the student in how chemical experiments are performed, how results are interpreted, and through its integrated laboratory sequence, to emphasize the importance of one subdiscipline in solving problems inherent to another. Meeting these goals, the Chemistry program provides the student with the flexibility to enter the industry, go on to graduate school, or pursue medical, dental, and other degrees in the health sciences.

Degrees

The Chemistry major may choose a program leading either to the BA or BS degree. The latter degree sequence has been approved by the American Chemical Society's Committee on Professional Training.

BA Program

The BA program offers the minimum fundamental knowledge required for adequate professional function in a career in chemistry. It is possible that students choosing this option may, through suitable use of unspecified semester credit hours, prepare for careers in areas as varied as chemistry-related businesses, government, medicine and dentistry, secondary school teaching, and even law or politics.

BS Program

The BS program provides more intensive training in chemistry for the student who intends either to obtain employment at the bachelor's level in the chemical industry or to pursue graduate study.

Bachelor of Arts in Chemistry

[Degree Requirements](#) (120 semester credit hours)

Faculty

FACG> nsm-chemistry-ba

Professors: Kenneth J. Balkus Jr., Inga H. Musselman, Bruce M. Novak, A. Dean Sherry, Mihaela C. Stefan, Jie Zheng

Associate Professors: Jung-Mo Ahn, Michael C. Biewer, Sheena D'Arcy, Gregg R. Dieckmann, Sheel Dodani, Jeremiah J. Gassensmith, Warren J. Goux, Gabriele Meloni, Steven O. Nielsen, Paul Pantano, John W. Sibert IV, Ronald A. Smaldone

Assistant Professor: Jiyong Lee

Professor Emeritus: Richard Caldwell

Professor of Instruction: Amandeep Sra

Associate Professors of Instruction: Sergio Cortes, Sandhya R. Gavva, Yu Huang, Yanping Qin

Assistant Professor of Instruction: Nimanka Panapitiya

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2417](#) Calculus I ^{1, 2}

or [MATH 2413](#) Differential Calculus ^{1, 2}

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[CHEM 1311](#) General Chemistry I ^{1, 3}

or [CHEM 1315](#) Honors Freshman Chemistry I ^{1, 3}

[CHEM 1312](#) General Chemistry II^{1, 3}_{_ _}

or [CHEM 1316](#) Honors Freshman Chemistry II^{1, 3}_{_ _}

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[MATH 2417](#) Calculus I^{1, 2}_{_ _}

or [MATH 2413](#) Differential Calculus^{1, 2}_{_ _}

[MATH 2419](#) Calculus II^{1, 2}_{_ _}

or [MATH 2414](#) Integral Calculus^{1, 2}_{_ _}

[PHYS 2125](#) Physics Laboratory I^{1, 3}_{_ _}

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat⁴_{_}

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 60-64 semester credit hours

Major Preparatory Courses: 28-31 semester credit hours beyond Core Curriculum

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I^{1, 3}

or [CHEM 1315](#) Honors Freshman Chemistry I^{1, 3}

[CHEM 1312](#) General Chemistry II^{1, 3}

or [CHEM 1316](#) Honors Freshman Chemistry II^{1, 3}

[CHEM 2323](#) Introductory Organic Chemistry I⁵

or [CHEM 2327](#) Honors Organic Chemistry I⁵

[CHEM 2325](#) Introductory Organic Chemistry II⁵

or [CHEM 2328](#) Honors Organic Chemistry II⁵

[CHEM 2233](#) Introductory Organic Chemistry Laboratory⁵

or [CHEM 2237](#) Honors Organic Chemistry Laboratory⁵

[NATS 1101](#) Natural Sciences and Mathematics Freshman Seminar

[CHEM 2401](#) Introductory Quantitative Methods in Chemistry

[PHYS 2325](#) Mechanics⁶

and [PHYS 2125](#) Physics Laboratory I^{7, 8}

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{6, 9}

[PHYS 2326](#) Electromagnetism and Waves

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

[PHYS 2126](#) Physics Laboratory II

MATH Sequence - Students may choose one of the following sequences:

I. [MATH 2413](#) Differential Calculus^{[1](#), [2](#)}

and [MATH 2414](#) Integral Calculus^{[1](#), [2](#)}

and [MATH 2415](#) Calculus of Several Variables

and [MATH 2418](#) Linear Algebra

or [STAT 2332](#) Introductory Statistics for Life Sciences

or

II. [MATH 2417](#) Calculus I^{[1](#), [2](#)}

and [MATH 2419](#) Calculus II^{[1](#), [2](#)}

and [MATH 3351](#) Advanced Calculus

and [MATH 2418](#) Linear Algebra

or [STAT 2332](#) Introductory Statistics for Life Sciences

Major Core Courses: 11 semester credit hours

[CHEM 3321](#) Physical Chemistry I

[CHEM 3471](#) Advanced Chemical Synthesis Laboratory

[CHEM 3472](#) Instrumental Analysis

Major Related Courses: 21-22 semester credit hours

Required Courses: 6-7 semester credit hours

[BIOL 3461](#) Biochemistry I

or [CHEM 3461](#) Biochemistry I

or [CHEM 4335](#) Polymer Chemistry

[CHEM 3341](#) Inorganic Chemistry I

or [CHEM 3322](#) Physical Chemistry II

Guided Electives: 15 semester credit hours

May be used in (partial) fulfillment of a Second Major, Minor or Teaching Certificate

III. Elective Requirements: 15-19 semester credit hours

- 17-18 semester credit hours needed if enroll in [STAT 2332](#) in Math Sequence (II. Major Requirements)
- 14-16 semester credit hours needed if enroll in [MATH 2418](#) in Math Sequence (II. Major Requirements)

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Bachelor of Science in Chemistry

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

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Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2417](#) Calculus I ^{1, 2} _{_}

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Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[CHEM 1311](#) General Chemistry I ^{1, 3} _{_}

or [CHEM 1315](#) Honors Freshman Chemistry I ^{1, 3} _{_}

[CHEM 1312](#) General Chemistry II ^{1, 3} _{_}

or [CHEM 1316](#) Honors Freshman Chemistry II ^{1, 3} _{_}

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Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

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Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[MATH 2417](#) Calculus I ^{1, 2} _{_}

or [MATH 2413](#) Differential Calculus ^{1, 2} _{_}

[MATH 2419](#) Calculus II^{1, 2}

or [MATH 2414](#) Integral Calculus^{1, 2}

[PHYS 2125](#) Physics Laboratory I^{1, 3}

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat⁴

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 62-66 semester credit hours

Major Preparatory Courses: 28-31 semester credit hours beyond Core Curriculum

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I^{1, 3}

or [CHEM 1315](#) Honors Freshman Chemistry I^{1, 3}

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or [CHEM 1316](#) Honors Freshman Chemistry II^{1, 3}

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or [CHEM 2327](#) Honors Organic Chemistry I⁵

[CHEM 2325](#) Introductory Organic Chemistry II⁵

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[NATS 1101](#) Natural Sciences and Mathematics Freshman Seminar

[CHEM 2401](#) Introductory Quantitative Methods in Chemistry

[PHYS 2325](#) Mechanics⁶ _{_}

and [PHYS 2125](#) Physics Laboratory I^{7, 8} _{_}

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat^{6, 9} _{_}

[PHYS 2326](#) Electromagnetism and Waves

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

[PHYS 2126](#) Physics Laboratory II

MATH Sequence - Students may choose one of the following sequences:

I. [MATH 2413](#) Differential Calculus^{1, 2} _{_}

and [MATH 2414](#) Integral Calculus^{1, 2} _{_}

and [MATH 2415](#) Calculus of Several Variables

and [MATH 2418](#) Linear Algebra

or [STAT 2332](#) Introductory Statistics for Life Sciences

or

II. [MATH 2417](#) Calculus I^{1, 2} _{_}

and [MATH 2419](#) Calculus II^{1, 2} _{_}

and [MATH 3351](#) Advanced Calculus

and [MATH 2418](#) Linear Algebra

or [STAT 2332](#) Introductory Statistics for Life Sciences

Major Core Courses: 11 semester credit hours

[CHEM 3321](#) Physical Chemistry I

[CHEM 3471](#) Advanced Chemical Synthesis Laboratory

[CHEM 3472](#) Instrumental Analysis

Major Related Courses: 23-24 semester credit hours

[CHEM 3322](#) Physical Chemistry II

[CHEM 3341](#) Inorganic Chemistry I

[BIOL 3461](#) Biochemistry I

or [CHEM 3461](#) Biochemistry I

[CHEM 4473](#) Physical Measurements Laboratory

[CHEM 4390](#) Research and Advanced Writing in Chemistry¹⁰

or [CHEM 4399](#) Research and Advanced Writing in Chemistry for Honors Students¹⁰

[CHEM 4V91](#) Research in Chemistry¹⁰ (3 semester credit hours)

[BIOL 3462](#) Biochemistry II

or [CHEM 3462](#) Biochemistry II

or [CHEM 4335](#) Polymer Chemistry

or [CHEM 4355](#) Computational Modeling

III. Elective Requirements: 13-17 semester credit hours

- 13-16 semester credit hours needed if enroll in [STAT 2332](#) in Math Sequence (II. Major Requirements)
- 12-15 semester credit hours needed if enroll in [MATH 2418](#) in Math Sequence (II. Major Requirements)

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

NOTE> Please be advised, the UTeach section below feeds in from a separate page updated by the UTeach team. Any changes made to the UTeach section below will not be retained.

UTeach Option

The [UTeach option](#) may be added to either the BA or BS degree. UTeach Dallas Option degree plans are streamlined to allow students to complete both a rigorous Bachelor of Science or Bachelor of Arts degree and all coursework for middle or high school teacher certification in four years. Teaching Option degrees require deep content knowledge combined with courses grounded in the latest research on math and science education. While most graduates go on to classroom teaching, UTeach alums are also prepared to enter graduate school and to work in a discipline related industry.

Fast Track Baccalaureate/Master's Degrees

For students with strong academic record who intend to pursue master's studies at UT Dallas, accelerated BS/MS Fast Tracks are available. After Fast Track admission to an MS program, students may take up to 15 semester credit hours of approved graduate courses in their senior year to use toward completion of both the BS and MS degrees. The available Fast Track options and their admission requirements, if any, in addition to having completed 90 or more semester credit hours (out of which 36 hours are from the core curriculum) with a cumulative GPA of at least 3.00, are as follows:

- [MS in Chemistry](#)
- [MS in Molecular and Cell Biology](#): Students must have a GPA of at least 3.20 and have completed [BIOL 2311](#), [BIOL 2312](#), and [BIOL 2281](#), or equivalent courses.
- [MS in Biotechnology](#): Students must have a GPA of at least 3.20 and have completed [BIOL 2311](#), [BIOL 2312](#), and [BIOL 2281](#), or equivalent courses.
- [MS in Bioinformatics and Computational Biology](#): Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.
- [MS in Data Science and Statistics](#): (Only Data Science and Applied Statistics specializations) Students must have a cumulative GPA of at least 3.20 in all mathematics and statistics courses.

Interested students, after reviewing the [UT Dallas Fast Track policy](#), should contact their undergraduate advisor and the graduate advisor of the intended MS program well in advance of their junior year to prepare a course sequence permitting maximal advantage and apply to the Fast Track program.

Minors

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

For all minors in the School of Natural Sciences and Mathematics students must complete all prerequisite sequences for required minor courses.

Minor in Chemistry

18 semester credit hours

Required: 11 semester credit hours

[BIOL 3461](#) Biochemistry I

or [CHEM 3461](#) Biochemistry I

[CHEM 3321](#) Physical Chemistry I

[CHEM 3472](#) Instrumental Analysis

Also:

A minimum of 7 additional semester credit hours in chemistry courses

1. A required Major course that also fulfills Core Curriculum requirements. If semester credit hours are counted in the Core Curriculum, students must complete additional coursework to meet the minimum requirement for graduation. Course selection assistance is available from the undergraduate advisor.
2. Three semester credit hours of Calculus are counted to fulfill the Mathematics Core Requirement with the remaining five semester credit hours to be counted under Component Area Option Core.
3. Six semester credit hours of Chemistry are counted under Science core, and one semester of Physics (PHYS 2125) is counted under Component Area Option core.
4. Students may use PHYS 2421 as part of the component area core curriculum requirement instead of PHYS 2125 if they choose that Physics sequence.
5. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
6. Students will take one of the two Physics sequences: PHYS 2325 and PHYS 2326 or PHYS 2421 and PHYS 2422 with accompanying labs.
7. A required Major course that also fulfills Core Curriculum requirement. Semester credit hours are counted in the Core Curriculum.
8. Six semester credit hours of Chemistry are counted under Science core, and one semester of Physics (PHYS 2125) are counted under Component Area Option core.
9. Students who complete PHYS 2421 do not need to complete PHYS 2125.
10. CHEM 4V91, CHEM 4390, and CHEM 4399 are better defined as a project than a course and constitute an important part of the BS degree. The student conducts original research under the supervision of a faculty member, and then must submit a research report which is defended orally. Normally this project will span two or more semesters. A complete set of guidelines is available from the undergraduate advisor.

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